

- 6 (a) all four diodes correct to give output, regardless of polarity
connected for correct polarity

M1
A1 [2]

(b) $N_S / N_P = V_S / V_P$
 $V_0 = \sqrt{2} \times V_{\text{rms}}$
ratio = $9.0 / (\sqrt{2} \times 240)$
= 1/38 or 1/37 or 0.027

C1
C1
A1 [3]

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7 (a) arrow pointing up the page B1 [1]

(b) (i) $Eq = Bqv$ C1
 $v = (12 \times 10^3) / (930 \times 10^{-6})$ C1
 $= 1.3 \times 10^7 \text{ m s}^{-1}$ A1 [3]

(ii) $Bqv = mv^2 / r$ C1
 $q/m = (1.3 \times 10^7) / (7.9 \times 10^{-2} \times 930 \times 10^{-6})$ C1
 $= 1.8 \times 10^{11} \text{ C kg}^{-1}$ A1 [3]

8 (a) momentum conservation hence momenta of photons are equal (but opposite) M1